

YETURI DINESH KRISHNA

☎ +91 83090 55501 — ✉ dineshkrishnayeturi@gmail.com — 🔗 LinkedIn — 🌐 GitHub — 📁 Portfolio

EDUCATION

Indian Institute of Technology (ISM) Dhanbad

B.Tech in Computer Science and Engineering CGPA: 7.20/10

Dhanbad, India

Expected June 2028

- **Relevant Coursework:** Data Structures & Algorithms, OOP with CPP & Java, Operating Systems, Database Management Systems, Theory of Computation, Computer Organization & Architecture

PROJECTS

Agent SLAM | Python, Asyncio, Google GenAI SDK, WebSockets, Pydantic | [GitHub](#) 2nd Place — IIT (ISM) Dhanbad

- Built a **Python WebSocket client** to maintain the debate transcript and handle server events, utilizing a custom **state machine** to track turn-based transitions.
- Integrated the **Google GenAI SDK** to generate context-aware counter-arguments by passing the real-time debate transcript directly into the prompt during the agent's turn.
- Implemented event-driven concurrency using Python **asyncio** to manage continuous WebSocket communication with the competition server while concurrently awaiting network responses from the **Gemini API**.
- Enforced structured JSON outputs from Gemini using **Pydantic schemas**, ensuring the model reliably returned its internal reasoning process and the final argument payload without parsing failures.

IntelliDesk AI | Next.js, TypeScript, Supabase, Pinecone, Google Gemini 2.5 Flash | [GitHub](#)

- Orchestrated an automated support pipeline using **Google Gemini 2.5 Flash** to efficiently classify incoming emails by issue category, severity, and customer sentiment.
- Prevented duplicate tickets by generating **768-dimensional embeddings** via Gemini and querying **Pinecone** for semantic similarity, combined with strict IMAP header parsing for accurate thread detection.
- Delivered real-time dashboard state synchronization using **Supabase PostgreSQL subscriptions**, implementing **P1-P4 SLA tracking** to visually alert support agents of impending response breaches.
- Engineered an intelligent auto-response system that asynchronously polls mailboxes via **IMAPFlow** and queries a Pinecone vector database of FAQs to draft context-aware support replies.

Rubik's Cube Solver | React Native, Three.js, Python, OpenCV, FastAPI, Zustand | [GitHub](#)

- Automated physical cube state extraction by building a **FastAPI** computer vision microservice, using **OpenCV** and **K-means clustering** in LAB color space to accurately classify colors across varying lighting conditions.
- Built a **60 FPS** interactive cross-platform 3D interface using **React Native** and **Three.js**, engineering a **quaternion based pivot grouping system** to smoothly rotate face slices without floating-point coordinate drift.
- Generated **sub-20-move solutions** in **under 100ms** by integrating Kociemba's Two-Phase algorithm, optimizing initialization with **asynchronous pruning tables** to prevent UI thread blocking.
- Managed application state and solve history using **Zustand + AsyncStorage**, ensuring reliable offline data persistence and supporting an active base of **30+ mobile users**.

TECHNICAL SKILLS

Languages: C++ (Proficient), Python, JavaScript, TypeScript, Java, SQL, HTML/CSS

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Asynchronous Programming, Database Management Systems (DBMS)

Frontend: React, Next.js, React Native (Expo), Tailwind CSS, Three.js, shadcn/ui, Figma, Canva

Backend & Databases: Node.js, FastAPI, PostgreSQL (PL/pgSQL), MongoDB, Supabase, Pinecone, Redis

AI & Tools: Google GenAI, OpenCV, Pydantic, Asyncio, WebSockets, Git, Linux, Vercel, PostHog, REST APIs

ACHIEVEMENTS & ACTIVITIES

- **Competitive Programming:** 2200+ Problems Solved | **LeetCode Knight** (1800+) | **CodeChef 3★** (1700+, Diamond Badge) | **Codeforces Pupil** (1300+)
- **Scholastic Excellence:** Secured **All India Rank 3012** in JEE Mains and **AIR 3348** in JEE Advanced (Top 0.2% among 1.5M+ candidates) | State Ranks **124** (AP) and **512** (TS) in EAMCET.
- **Hackathons & Competitions:** Engineered complex full-stack architectures under tight time constraints, winning **2nd Place** at Agent SLAM (IIT ISM) and successfully deploying IntelliDesk AI at **HackOdisha**.
- **Fellowships & Programs:** Ranked in the **Top 8% Nationwide** for the AlgoUniversity Tech Fellowship 2025 | Completed the Google Student Upskilling Program focusing on advanced DSA strategies.